

GOOGLE PLAY STORE DATA ANALYTICS

Project Report

Submitted as part of: ElevanceSkills Internship / Project Program

Project Title: Google Play Store Data Analytics

Technology Used: Python, Pandas, NumPy, Matplotlib, Seaborn

Dataset Used: Google Play Store Dataset and User Reviews Dataset

1. ABSTRACT

The **Google Play Store Data Analytics** project focuses on analyzing applications available on the Google Play Store to understand their characteristics, popularity, ratings, reviews, installations, pricing, and other important attributes.

The project uses Python-based data analytics techniques to clean, transform, analyze, and visualize the available data. The analysis helps identify patterns and relationships between different application attributes such as category, rating, reviews, installs, price, size, and type.

As part of the project, multiple analytical tasks and visualizations were created to present the data in an understandable and meaningful way. These visualizations help users identify trends, compare application categories, understand geographical and time-based patterns, and analyze relationships between important application metrics.

2. INTRODUCTION

The Google Play Store contains a large number of applications belonging to different categories such as Education, Entertainment, Games, Business, Health & Fitness, Photography, Finance, and many others.

Analyzing this data can provide useful insights into:

- Which application categories are popular
- How application ratings are distributed
- The relationship between reviews and installations
- Differences between free and paid applications
- Application pricing patterns
- Changes and trends over time
- Relationships between multiple application characteristics

Python provides several libraries that make it possible to perform these analytical operations efficiently. In this project, Python and its data analysis libraries were used to clean the datasets, perform exploratory analysis, and create visualizations.

3. OBJECTIVES

The main objectives of this project are:

1. To understand and analyze Google Play Store application data.
 2. To clean and prepare the dataset for analysis.
 3. To perform exploratory data analysis using Python.
 4. To analyze application categories, ratings, reviews, installs, size, price, and type.
 5. To identify relationships between different application attributes.
 6. To create meaningful visualizations from the analyzed data.
 7. To complete the six assigned analytical tasks.
 8. To present the findings through clear and interactive data visualizations.
-

4. DATASETS USED

Two datasets were used for the project:

4.1 Play Store Data

File: Play Store Data.csv

This dataset contains information about applications available on the Google Play Store.

Important columns include:

- App
- Category
- Rating
- Reviews
- Size
- Installs
- Type

- Price
- Content Rating
- Genres
- Last Updated
- Current Ver
- Android Ver

4.2 User Reviews Data

File: User Reviews.csv

This dataset contains user review information associated with applications.

Important fields include:

- App
- Translated_Review
- Sentiment
- Sentiment_Polarity
- Sentiment_Subjectivity

These datasets provide both application-level information and user feedback that can be used for deeper analysis.

5. TECHNOLOGIES AND LIBRARIES USED

Programming Language

Python

Libraries

Pandas: Used for loading, cleaning, transforming, and analyzing datasets.

NumPy: Used for numerical operations.

Matplotlib: Used to create analytical charts and visualizations.

Seaborn: Used to create statistical and analytical visualizations.

Jupyter Notebook: Used as the development and analysis environment.

6. DATA PREPROCESSING

Before performing the analysis, the datasets were examined and cleaned.

The major preprocessing activities included:

- Loading the CSV datasets.
- Checking the structure of the datasets.
- Identifying missing values.
- Handling duplicate records.
- Cleaning numerical columns.
- Converting values into appropriate data types.
- Processing application size and price values.
- Processing installation values.
- Converting date information into a suitable format.
- Preparing the cleaned data for visualization and analysis.

Data preprocessing was necessary to ensure that the visualizations and calculations were based on consistent and usable data.

7. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the Play Store dataset.

The analysis examined:

- Application categories
- Application ratings
- Number of reviews
- Number of installations
- Free versus paid applications
- Application prices
- Application sizes
- Content ratings
- Genres

- Last updated dates
- User sentiment information

The EDA helped identify patterns and relationships that were later represented through the assigned visualization tasks.

8. ANALYTICAL TASKS

TASK 1 – CATEGORY-BASED ANALYSIS

A grouped bar chart was created to compare relevant application metrics across different Google Play Store categories.

The visualization allows different categories to be compared side by side and helps identify differences in application performance.

Purpose

The objective of this task was to provide a clear category-level comparison using the available Play Store metrics.

Visualization

Grouped Bar Chart

The chart provides an easy comparison between categories and helps identify categories with relatively higher or lower values for the selected metrics.

TASK 2 – GEOGRAPHICAL ANALYSIS

A geographical visualization was created to represent application-related information across different geographical regions.

Purpose

The objective of this task was to present geographical patterns using a map-based visualization.

Visualization

Choropleth Map

A choropleth visualization represents values using different levels of intensity across geographical areas.

This provides a visual way of identifying geographical differences in the analyzed data.

TASK 3 – COMPARATIVE METRIC ANALYSIS

A dual-axis visualization was created to compare two different metrics within the same analytical view.

Purpose

The purpose of this task was to examine the relationship and variation between two metrics that may have different scales.

Visualization

Dual-Axis Chart

Using two axes allows both metrics to be displayed together while maintaining their individual scales.

This makes it easier to observe changes and possible relationships between the selected measures.

TASK 4 – TIME-SERIES ANALYSIS

A time-series visualization was created to analyze changes in the selected Play Store metrics over time.

The date-related information was processed before creating the visualization.

Purpose

The objective was to understand how application-related activity or metrics changed over different time periods.

Visualization

Time-Series Chart

The visualization makes it easier to observe increases, decreases, fluctuations, and general patterns over time.

TASK 5 – RELATIONSHIP ANALYSIS

A bubble chart was created to analyze relationships between multiple application-level variables.

Purpose

The objective was to represent multiple dimensions of the dataset within a single visualization.

Visualization

Bubble Chart

A bubble chart allows values to be represented using:

- X-axis
- Y-axis
- Bubble size
- Category/group

This provides a multidimensional view of the selected Play Store metrics and helps identify patterns between applications.

TASK 6 – TREND AND COMPOSITION ANALYSIS

A stacked area chart was created as the final analytical visualization.

Purpose

The objective was to analyze how different groups contributed to the overall value and how their contribution changed over time.

Visualization

Stacked Area Chart

The stacked area chart provides a visual representation of:

- Changes over time
- Contribution of different groups
- Overall trends
- Relative composition

This visualization provides a broader view of how the selected categories or metrics evolved during the analyzed period.

9. DASHBOARD AND VISUALIZATION

The completed project combines the analysis and assigned visualizations into a structured analytical presentation.

The dashboard/project visualization includes the outputs from the six assigned tasks:

1. Grouped Bar Chart
2. Choropleth Map
3. Dual-Axis Chart
4. Time-Series Chart
5. Bubble Chart
6. Stacked Area Chart

The visualizations were designed to make the results easier to understand compared with viewing raw tabular data.

10. KEY INSIGHTS

The analysis provides several useful observations about Google Play Store applications.

Application Categories

The Play Store contains applications across a wide range of categories, and the distribution of applications is not uniform across categories.

Ratings

Application ratings provide an indication of user satisfaction, although ratings should be considered together with other metrics such as review count and installation numbers.

Reviews and Installs

Applications with a larger number of reviews generally represent applications with greater user interaction. Comparing reviews and installations helps understand application engagement.

Free and Paid Applications

The dataset contains both free and paid applications. Comparing these types helps understand differences in application availability and pricing.

Application Pricing

Paid applications show variation in pricing, allowing pricing patterns to be explored across categories.

Time-Based Patterns

The time-series analysis provides an understanding of how selected application-related metrics vary over time.

User Sentiment

The User Reviews dataset provides additional information about user opinions through sentiment, polarity, and subjectivity values.

11. BENEFITS OF THE PROJECT

This project demonstrates how raw application data can be transformed into meaningful information through data analytics.

The project provides experience in:

- Data cleaning
 - Data preprocessing
 - Exploratory Data Analysis
 - Python programming
 - Pandas and NumPy
 - Data visualization
 - Statistical analysis
 - Business-oriented interpretation of data
 - Creating analytical dashboards and charts
-

12. CHALLENGES FACED

During the project, several data-related challenges were addressed, including:

- Missing values in the dataset
- Duplicate application records
- Inconsistent numerical formats
- Different representations of installation values
- Data type inconsistencies
- Preparing date fields for time-based analysis

- Converting raw data into suitable formats for visualization

These challenges helped improve understanding of real-world data preprocessing and analytics.

13. CONCLUSION

The **Google Play Store Data Analytics** project demonstrates the complete process of converting raw application data into meaningful analytical insights.

Using Python, Pandas, NumPy, Matplotlib, and Seaborn, the project involved data preprocessing, exploratory analysis, and the creation of multiple visualization techniques.

The six analytical tasks provided different perspectives of the dataset through category-based, geographical, comparative, time-series, relationship, and composition-based visualizations.

Overall, the project helped develop practical skills in **Python-based data analysis and visualization** and provided experience in working with a real-world dataset.

14. FUTURE SCOPE

The project can be further extended by:

- Adding more advanced statistical analysis.
 - Applying machine learning techniques to predict application ratings or popularity.
 - Performing deeper sentiment analysis of user reviews.
 - Creating interactive dashboards.
 - Adding additional filters and analytical parameters.
 - Studying application success factors.
 - Developing predictive models for application performance.
-

15. PROJECT OUTCOME

The final outcome of the project is a structured **Google Play Store Data Analytics solution** containing:

- Cleaned and processed datasets

- Exploratory data analysis
- Python-based analytical code
- Six analytical visualizations
- Dashboard/report presentation
- Business-oriented observations and insights

The project demonstrates the practical application of data analytics techniques to a real-world dataset and provides a foundation for further data analysis and visualization projects.

16. PROJECT SUBMISSION

Project Name: Google Play Store Data Analytics

Dataset: Google Play Store Data + User Reviews

Programming Language: Python

Development Environment: Jupyter Notebook

Main Libraries: Pandas, NumPy, Matplotlib, Seaborn

Completed Components: Tasks 1–6, data preprocessing, analysis, and visualization

Project Status: Completed